

07 May 2026 (BI Engineer) -> Data slave

Data

Product: Analytics (Dashboards, Table, Reports)
Data Eng (Data Pipelines and Data Infra),
Data science (Models)

Data -> data is information that is structured and has sense

1. Information
2. Data
3. Knowledge
4. Wisdom

Database -> For daily operations
A database is used to store and manage current, live data that applications use everyday.

Data Warehouse -> For analysis & Reporting

A data warehouse is used to analyze large amounts of historical data to help businesses make decisions.

- Data Engineer - (Data Architecture, ETL Per, Data Ops)
- Data Scientist - (AI Engineer, ML Ops Engineer)
- Data Analyst - (Power BI Dev, Computational Intelligence Analyst)

11 May 2026 →

Case statements

→ Allows us to build logical conditions

Syntax:

SELECT Column 1,
CASE

WHEN Condition 1 THEN 'Result 1'

WHEN Condition 2 THEN 'Result 2'

ELSE 'default result'

END AS Column-name
Column 2,
Column 3

FROM table-name

we build conditions based on other columns

default result comes out when the conditions are not met

→ A case statement creates a new column

→ A case statement is found within a select statement

Example (Exercise) SQL Case Statement

Table: products

Product-Id	Product-name	Price
1	Laptop	1200.00
2	Phone	800.00
3	keyboard	45.00
4	Monitor	300.00
5	Mouse	25.00

2. Query:

- A case statement is a way of building a categorical column using a numerical column.

```
SELECT product_name,  
       price,  
  
       CASE  
         WHEN price > 1000 THEN 'Expensive'  
         WHEN price BETWEEN 100 AND 1000 THEN 'Mid-range'  
         WHEN price < 100 THEN 'Budget'  
       END AS price-category  
FROM products;
```

↓ output-

product-id	product-name	price	price category
1	Laptop	1200	Expensive
2	Phone	800	Mid-range
3	Keyboard	45	Budget
4	Monitor	300	Mid-range
5	Mouse	25	Budget

2. Table : orders

order-id	Customer-name	Amount
1	Alice	150.00
2	Bob	560
3	Charlie	999.99
4	Diana	45.50
5	Ethan	1200.00

Question

Label each order

- High Value
- Medium Value
- Low Value

Query

```
SELECT Customer-name,  
        amount  
CASE  
    WHEN amount >= 1000 THEN 'High Value'  
    WHEN amount BETWEEN 500 AND 999.99  
        THEN 'Medium Value'  
    WHEN amount < 500 THEN 'Low Value'  
END AS order-value-category  
FROM orders ;
```

↓ output

Customer-name	amount	order-value-category
Alice	150.00	Low Value
Bob	560.00	Medium Value
Charlie	999.99	Medium Value
Diana	45.50	Low Value
Ethan	1200.00	High Value

My attempts →

2. Table : products

Q : classify each Product by Price

- 'expensive' if price > 1000
- 'Mid-range' if price between 100 and 1000
- 'Budget' if price < 100

Answer:

```
SELECT price,  
CASE  
    WHEN price > 1000 THEN  
    WHEN price BETWEEN 100 AND 1000 THEN  
    WHEN price < 100 THEN  
ELSE END AS '
```